

18 April 88

Fourteenth Annual In-House Convention

25-26 April 1988

Room 4258/63

Department of Psychology

Dalhousie University

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Schedule

Monday, 25 April

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|--------------|----------------|--|
| 12:45 - 2:15 | Session I | 1. I.A. Meinertzhagen & A. Ferrús |
| | Chair: M. Yoon | 2. S. Shaw, D. Moore, A. Fröhlich,
J. Dawson & P. Dickson |
| | | 3. B. Rusak, R. Mistlberger, B.
Losier & C.H. Jones |
| | | 4. E. Coscia |
| | | 5. D.P. Phillips & J.B. Kelly |

COFFEE BREAK

- | | | |
|-------------|--------------------|----------------------------------|
| 2:30 - 3:45 | Session II | 6. W.K. Honig & K. Stewart |
| | Chair: J. Connolly | 7. A. Droungas & V.M. LoLordo |
| | | 8. A.R. Delamater & V.M. LoLordo |
| | | 9. S. Nakajima |

COFFEE BREAK

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| 4:00 - 5:00 | Session III | 10. R. Rodger |
| | Chair: N. Wadden | 11. T. Juckes |
| | | 12. J. Fentress |
| | | 13. J. Barresi |

BEER IN LOUNGE

Tuesday, 26 April

- | | | |
|--------------|-----------------|---|
| 9:45 - 10:00 | Session IV | 14. P. Dunham, F. Dunham & C. Davis |
| | Chair: R. Croll | 15. E. Walsh, M. Legerstee, & K.
Kienapple |
| | | 16. K. Baird & A. Cohen |
| | | 17. A. Cohen |

ABSTRACTS

1. I.A. Meinertzhagen and A. Ferrús*. *C.S.I.C., Velazquez 144, 28006 Madrid, Spain.

The Influence of Cell Size Upon Synaptic Frequency in Drosophila.

The size of a neuron provides a major prediction of both the number of its synapses and of its aggregate area of synaptic contact. Our previous studies establish a close correlation between the number of afferent synaptic sites and the presynaptic membrane surface area over which these are distributed, in the terminals of photoreceptors innervating the first optic neuropile, or lamina, underlying the fly's eye. A similar relationship also exists for their target interneurons, L1 and L2, each unique occupants of one of the cartridges of lamina neuropile, the dendrites of which also receive synaptic input in proportion to their surface area. All such studies are correlative, however, and fail to distinguish directly whether large neurons bear more synapses or, alternatively, whether neurons with more synapses grow larger, perhaps for some trophic reason.

I will describe experiments in Drosophila in which the causal relationship between cell size and synaptic number may be established by analysing synaptic frequencies in neurons which are made independently large, using genetic mutants.

2. S. Shaw, D. Moore, A. Fröhlich, J. Dawson and P. Dickson, The Fly Collective, Psychology, Dalhousie University, and Biology, MSVU.

La nouvelle sensation: plus ça change, plus ce n'est pas la même mouche.*
(*Supported by NSERC and a rather small dictionary).

The nature of evolutionary change has been little studied at the cellular level in any nervous system, and so constitutes an important lacuna in knowledge. We have told you before that in the system we study for convenience, in insects, evidence suggests that the evolution of function has been accomplished by saving the same fixed set of ancestral neuronal homologues, but allowing these to vary some of their mutual synaptic connections. Attempts to justify a statement of such penetrating simplicity will be made with the help of actual results, drawn from our current work on a particularly stupid and probably primitive fly. The new results throw more light on the things that may change more readily, and things that may not, in the evolution of a nervous system.

3. B. Rusak, R. Mistlberger, B. Losier and C.H. Jones

The Food, the Pad and the Cuddlies.

Rats and a few other species have been shown to anticipate a daily time at which they are fed by increasing their activity before this time. This increased anticipatory activity has been shown to be under the control of a food-entrainable oscillator (FEO), a component of the circadian system. This oscillator is separate from the light-entrainable pacemaker in the

obtained. Numerosity is established as a "robust" stimulus dimension. The mechanism is a mystery, but the adaptive value of the discrimination would be considerable for a foraging animal.

7. A. Droungas and V. LoLordo

Short Exposure to Long Noises in the Midst of Shocks Will Make Rats Relax: Reasons Remain to be Determined.

It is well documented that a CS, like a noise or a light, that is presented on trials during which shock is postponed will become a fear inhibitor. In the presence of such a CS, which has been "explicitly unpaired" with shock, rats will continue to lever-press for food even if another CS that has been "explicitly paired" with shock is concurrently present. The present experiment demonstrated that such fear inhibiting property was acquired by the CS after what is typically considered "limited" amount of training (8 conditioning sessions) but only if the explicitly unpaired CS was of relatively long duration (60 s). An identically trained CS of short duration (10 s) failed to become a fear inhibitor.

8. A. R. Delamater and V. M. LoLordo

Images, Expectancies, and Attributions: A Levels-of-Processing Approach to Pavlovian Learning.

Suppose the rat was exposed repeatedly to a tone followed by sugar water, until the tone elicited the response of approach to the food source. If we then, without the tone, let the rat drink sugar water but force illness upon him afterwards, we see that he subsequently avoids drinking the sugar water. What would happen if we then present the tone again in the absence of sugar water? In another scenario, suppose that after the tone-sugar pairings we repeatedly present the tone but this time withholding sugar water until the approach response is no longer evoked. If sugar water is then brought back to the situation, what would happen if we then re-present the tone? In yet a third scenario, suppose that we pair simultaneously a tone and a light with the availability of sugar water. After this we present the light without also presenting sugar water. Does the rat become more strongly convinced that tone was the cause? Results from experiments such as these suggest that different "fundamental" mechanisms of Pavlovian learning may emerge at different levels within the system.

9. S. Nakajima

Blockade of Dopamine D2 Receptors in the Brain Suppresses Operant Responding.

Raclopride, a dopamine antagonist, has a high affinity for D2 receptors but extremely low affinity for D1 receptors. Intraperitoneal injection of raclopride suppressed bar-pressing for food (CRF and VI-30 sec), ventral tegmental stimulation, and intravenous heroin infusion in the rat. The degree of suppression was related to the dose, and the maximal effects were observed from 30 to 180 min after injection. Raclopride injection into the nucleus accumbens suppressed ipsilateral tegmental self-

roles of experience, b) rules of generalization, and c) the organism's handling of incomplete or distorted information. This talk is designed to highlight some of these issues, with reference to existing literature and potential experimental approaches for future work. Naturally, mice and wolves will play a key (but not exclusive) role here!

✓ 13. J. Barresi

The Myriad Mind: Steps Toward a Science of Persons.

Psychologists have given insufficient attention to the possible use of biographical materials involving whole lives as a data base for the development of the psychology of persons. In a book that I am currently working on, I will argue for an approach for using such data which treats individual lives as a unit. According to this approach, competing theories would be developed to account for the causal history of particular lives, and the power of these theories would be assessed relative to alternative theories in terms of their simplicity, coherence, and plausability in accounting for the particular life, and in terms of the generality and testability of their principles in applying to other lives.

14. P. Dunham, F. Dunham and C. Davis

Mother Care and Other Care: Some Differences in Mother-Infant Social Structure.

While studying the structure of mother-infant social interactions, we encountered some unexpected and interesting differences when comparing mothers employed outside the home and mothers who care for their infants full-time in the home.

Poster 15. E. Walsh, M. Legerstee and K. Kienapple

The Development of Imitation During the First Year.

Many studies have looked at imitation in infants under constrained, unnatural conditions and at a particular age. In contrast, this study has taken a longitudinal approach to study the development of imitation by infants when engaged in natural interactions with their mothers. Play sessions of 2 mothers and their infants were videotaped. Dyads were observed biweekly from 3 to 25 weeks of age and monthly thereafter until 12 months. Sessions were of 45 sec durations at 3, 5 and 7 weeks and of 60 sec thereafter. Patterns of imitation (especially in mouth opening, lip movements, smile and vocalization) indicate that an imitative ability may be present in the first 3 to 9 weeks of an infant's life. Imitation appears to decrease at about 17 weeks and increase again at about 34 to 40 weeks. The results suggest that imitation may be an integral part of early mother/infant social interactions.

19. R. Klein and C. Smith

Does Covert Visual Orienting Affect Stimulus Encoding or Decision Criteria?

The debate goes on. Investigators have applied paradigms from within mental chronometry, signal detection theory and electrophysiology to answer this question. This literature will be critically reviewed with particular emphasis on the distinction between endogenous and exogenous covert orienting. Mathematical frameworks like the accumulator and random walk models will be applied to the corpus of chronometric data from the Hansen thesis and similar follow-up experiments conducted in our laboratory. By focussing attention on a detailed analysis of these studies we hope to enhance the "signal" in this noisy area. Strategies for future research will be presented.

20. K. Briand

Implicit Expectancies in Automatic Visual Orienting.

Peripheral visual stimuli can attract attention very quickly, and such orienting effects appear to be automatic. It is usually assumed that the cue operates via a system which informs higher-level processes 'where' in space an event occurs, and that 'what' is at the cued location is irrelevant. From this it follows that (all other things being equal) a given cue will have identical effects on all possible classes of targets which occur. This is not the case. Cueing effects are modulated by the physical similarity between the cue and target, indicating that when a peripheral event occurs, information concerning attributes such as form, orientation, and motion is used to facilitate processing of subsequent targets. Color information does not appear to be used in this manner. It seems that when one orients attention to a location via some peripheral 'event', one expects to find the object responsible for that 'event'.

21. G. Eskes, S. Bryson and T. McCormick

Further Studies of Comprehension in Autistic Children.

Converging evidence from our studies and those from other laboratories suggest that high-functioning autistic children's comprehension of single words is similar to that of carefully matched control children. However, while single word comprehension may be developmentally normal, both clinical reports and language comprehension studies agree that autistic children do have trouble comprehending more complex, sequentially ordered information, such as sentences or paragraphs. This talk will describe a study we are about to begin that will examine their understanding of the same words presented individually or in the context of a sentence.

22. A. Cohen

To Tune or Not to Tune: Cognitive and Sensory Influences on Memory for a Microtonal Scale as Shown by a Signal Detection Analysis.

Musically trained and less trained subjects completed nine conditions of an absolute judgment task resulting from three sequential (melodic) and

between people and objects, they do not seem to differentiate between the active and inactive person as early and as clearly as do normal infants. Rather than showing distress, DS babies continue to smile and vocalize with neutral faces when in front of the nonresponsive adult. This lack of communication skills may be one of the contributing factors to their delayed linguistic development. Additional results focused on the evolution of the responses to people and objects of these DS infants during their first year of life.

25. J. Ryon and R.E. Brown

Urine-Marking in Female Wolves (*Canis lupus*): An Indicator of Dominance Status and Reproductive State.

Urine marking was observed for over 5 years in a pack of wolves (6 females and 9 males) in a semi-natural enclosure. The dominance status of each pack member was determined by body postures during spontaneous interactions. Urine-marking was most frequent by the alpha female and much less frequent by subdominant females. All pack members sniffed the urine marks of all females. Males licked the urine marks more than females and urine-licking by both males and females increased in the winter mating season, ceasing abruptly at the end of the females' estrus in March. Most over-marking was done by the alpha male and female. The frequency of over-marking decreased while urine-licking increased as the females entered estrus. We can conclude that urine-marking by female wolves gives information on dominance status and reproductive state. Investigation of urine-marks, particularly by licking, may help to regulate mating behaviour in wolves in both sexes.

26. L. Wisker

Effects of Genetic Differences at the Major Histocompatibility Complex and Diet on Attraction to the Maternal Odour by H-2 Congenic Mice.

Mouse pups are attracted to the "maternal odors" produced by lactating females. This experiment investigates the role of diet and the Major Histocompatibility Complex (MHC) in maternal odor recognition. Two congenic strains of mice, which differ in the MHC were used (C57BL/6J and B6C-H-2^{bml}/BYJ). Pups were tested for odor preferences using maternal urine and feces odors in 2 min. tests. If the MHC plays an important role in maternal odor identification, odors from the same strain should be preferred to those of the other strain. If diet influences odor identification, odors from mothers on the same diet should be preferred. Finally, it is possible that the MHC will influence preferences for urine ~~or~~ while diet influences preferences for feces odors.

27. B.J. Chiasson and R.P. Croll

Catecholamines in the Central Nervous System of *Lymnaea*.

The distribution of catecholamines (CA's) within the central ganglia of the pond snail, *Lymnaea stagnalis* was examined through the use of glyoxylic acid and aqueous formaldehyde histofluorescence techniques. We found prominent clusters of CA cells located in the right and left parietal ganglia and along the anteriolateral margin of the pedal ganglia.

Male Parental Behaviour in the House Mouse (*Mus musculus*).

All studies of paternal behaviour have been done with ad lib food and water in a warm environment. No studies have examined the importance of the male when environmental conditions mimic the real world, i.e. food must be foraged. Two groups of house mice were compared in a controlled environment: litters with their dam alone and litters with both parents. The environment consisted of a home cage with a running wheel and a feeder controlled by a computer. The amount of running needed to obtain food was manipulated. The growth and development of the pups was monitored by measuring the number of pups/litter, pup weight, growth rate, eye and ear opening, the number of pups at weaning and the weight of pups at weaning. Differences on these measures between the two groups was determined. Behavioural observations were also conducted to examine the male's contribution to litter development.